Chapter 3. High-Dimensional Data and Regularization

Predictive Modeling, Lecture 2 (OLS instability, Ridge, Lasso, Elastic Net, choosing λ)

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1 Linear Prediction and Why It Becomes Unstable

1.1 Setting

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad X \in \mathbb{R}^{n\times p}, \qquad \hat{y} = x_{\text{new}}^{\top}\hat{\boldsymbol{\beta}}.$$

n is the number of observations, p the number of candidate variables, β_j the linear contribution of variable j. Regularization selects, among all $\hat{\boldsymbol{\beta}}$ that fit the observed $\mathbf{y}$ well, a solution that is *less sensitive* on new x. That a coefficient is stable does not mean a causal effect has been identified.

1.2 High dimension is about p relative to n

Situation	Example	Problem for prediction
$p \ll n$	100,000 customers, 20 variables	fairly stable, but collinearity is still possible
p close to n	500 patients, 400 variables	small data changes move coefficients and selection a lot
$p \geq n$	1,000 documents, 50,000 words	OLS is not unique or cannot be computed
many derived features	raw variables, interactions, one-hot encodings	search degrees of freedom grow faster than signal

1.3 OLS and what it assumes

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 = (X^{\top}X)^{-1}X^{\top}\mathbf{y} \quad \text{(if } X^{\top}X \text{ is invertible)}.$$

The formula requires $X^{\top}X$ to be well conditioned, i.e. the variables must provide reasonably independent directions of information. From a predictive point of view, OLS is not a "wrong model" but a baseline that *can become unstable.*

1.4 When $p \geq n$: non-uniqueness

Theorem: interpolation is not unique

If $\text{rank}(X) < p$ (always the case when $p > n$), there is $v \neq 0$ with $Xv = 0$, and then $X(\hat{\boldsymbol{\beta}} + v) = X\hat{\boldsymbol{\beta}}$ for every solution $\hat{\boldsymbol{\beta}}$. In particular, when $\text{rank}X = n$, infinitely many $\boldsymbol{\beta}$ give $X\boldsymbol{\beta} = \mathbf{y}$ with zero training error.

The direction v does not change training predictions, but on a new case $x_{\text{new}}^{\top}v$ need not be 0: zero training error says nothing about which solution predicts well. Regularization deliberately gives up some of these degrees of freedom. (Among all interpolators, the minimum-norm one $X^{\top}(XX^{\top})^{-1}\mathbf{y}$ is the $\lambda \to 0$ limit of Ridge.)

1.5 Collinearity and the variance of coefficients

Income, credit limit and monthly card spending may all measure the same customer dimension: one coefficient may become very positive and another very negative; the combined prediction is similar but the individual coefficients wobble and their roles can be reallocated in a new sample.

Theorem: variance along eigen-directions

Under $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$, $\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}$. With the SVD $X = UDV^\top$, $D = \mathrm{diag}(d_1, \ldots, d_p)$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 \sum_{j=1}^{p} \frac{v_j v_j^\top}{d_j^2}, \qquad \mathrm{Var}(v_j^\top \hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{d_j^2}.$$

Proof. $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$, so the covariance is $\sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}$. Since $X^\top X = V D^2 V^\top$, its inverse is $V D^{-2} V^\top$. ■

If some direction of $X^\top X$ is nearly 0 ($d_j \approx 0$), the inverse explodes and the coefficient variance in that direction blows up. Equivalently, a small data perturbation is amplified: $\Delta\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top \Delta \mathbf{y}$. In practice: CV averages look similar while fold-wise coefficients, selected variables and prediction rankings change a lot. Check the stability of λ and of the selected variables, not only performance.

2 Regularization: Buying Variance With Bias

$$\mathbb{E}[(Y - \hat{f}(X))^2] = \mathrm{Bias}^2 + \mathrm{Var} + \sigma^2.$$

OLS aims at low bias; in high dimensions the variance term can be large. Regularization steps back from the training optimum and reduces the size of coefficients, the degrees of freedom, and sample sensitivity. Success is judged by CV and the final test, not by training loss. Low regularization tracks noise in the training data, adequate regularization is a candidate for minimal generalization risk, excessive regularization shrinks even the signal.

2.1 One template, three models

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda P(\boldsymbol{\beta}) \Big], \qquad P(\boldsymbol{\beta}) = \begin{cases} \|\boldsymbol{\beta}\|_2^2 & \text{Ridge} \\ \|\boldsymbol{\beta}\|_1 & \text{Lasso} \\ \alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2 & \text{Elastic Net} \end{cases}$$

The first term is data fit, the second is model complexity. $\lambda = 0$ gives OLS; larger λ means stronger shrinkage and simplification. The good λ is not the one with the smallest training loss but the one with the smallest loss on new data.

2.2 Standardization comes first

$$x_{ij}^{\mathrm{std}} = \frac{x_{ij} - \mu_j^{(\mathrm{train})}}{s_j^{(\mathrm{train})}}.$$

A penalty on $|\beta_j|$ depends on the unit of X_j (card spending in won versus number of clicks): rescaling X_j by c rescales its unpenalised coefficient by $1/c$ and so changes how hard it is penalized. Standardization makes the penalty unit-free. The intercept is usually not penalized (center y and X first). One-hot variables, sparse matrices and already meaningful units need separate judgement. Crucially, μ_j and s_j are estimated **on the training part only**.

3 Ridge

3.1 Closed form and constrained form

$$\hat{\boldsymbol{\beta}}_\lambda^{\text{Ridge}} = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2\right] = (X^\top X + n\lambda I)^{-1} X^\top \mathbf{y}.$$

Proof. The gradient is $-\frac{2}{n}X^\top(\mathbf{y} - X\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta}$; setting it to zero gives $(X^\top X + n\lambda I)\boldsymbol{\beta} = X^\top \mathbf{y}$. The matrix is positive definite for every $\lambda > 0$, so the solution is unique even when $p \geq n$, and the objective is strictly convex. ■

Large coefficients cost more; most coefficients approach 0 but are not exactly 0. Ridge is natural when the predictive signal is spread over many variables. By Lagrangian duality the problem is equivalent to $\min \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$ subject to $\|\boldsymbol{\beta}\|_2^2 \leq c$, for some $c = c(\lambda)$.

3.2 Ridge shrinks small singular directions most

Theorem: SVD form of Ridge

With $X = UDV^\top$ and $\tilde{\lambda} = n\lambda$ (the slides absorb n into λ),

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = V\,\text{diag}\Big(\frac{d_j}{d_j^2 + \tilde{\lambda}}\Big)U^\top \mathbf{y}, \qquad X\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \sum_j u_j \frac{d_j^2}{d_j^2 + \tilde{\lambda}} u_j^\top \mathbf{y}.$$

Proof. $X^\top X + \tilde{\lambda} I = V(D^2 + \tilde{\lambda} I)V^\top$ and $X^\top \mathbf{y} = VDU^\top \mathbf{y}$, so $(X^\top X + \tilde{\lambda} I)^{-1} X^\top \mathbf{y} = V(D^2 + \tilde{\lambda} I)^{-1} DU^\top \mathbf{y}$. ■

OLS multiplies direction j by $1/d_j$; Ridge by $d_j/(d_j^2 + \tilde{\lambda})$, which stays bounded by $1/(2\sqrt{\tilde{\lambda}})$. Directions with small d_j are those the data can hardly distinguish, and Ridge shrinks exactly those most strongly. The **effective degrees of freedom** $\text{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \tilde{\lambda})$ decrease from p to 0 as λ increases. Numerical stability, coefficient stability and prediction on new data can improve simultaneously.

3.3 Ridge shares signal among correlated variables

Theorem: identical columns get identical Ridge coefficients

If $x_{(1)} = x_{(2)}$, then $\hat{\beta}_1^{\text{Ridge}} = \hat{\beta}_2^{\text{Ridge}}$ for every $\lambda > 0$.

Proof. The fit term depends on β_1, β_2 only through $\beta_1 + \beta_2$. Swapping $\beta_1 \leftrightarrow \beta_2$ leaves the objective unchanged, and the objective is strictly convex, so the unique minimizer must be symmetric. ■

OLS can produce opposite-sign coefficients such as $\hat{y} \approx x_1(12) + x_2(-11)$; the sum predicts similarly but each coefficient is unstable across resamples. Ridge tends towards $\hat{y}_{\text{Ridge}} \approx x_1(0.8) + x_2(0.7)$: it shrinks similar variables together so that no single one takes an excessive role (illustrative, not literal numbers). If stable prediction rather than variable selection is the goal, Ridge is a strong starting point.

Data / goal	Why Ridge fits	What to check
many weak signals	keeps all coefficients a little	does it beat OLS?
strong multicollinearity	stabilizes unstable directions	standardization and coefficient paths
$p \geq n$ or sparse matrices	unique, computable solution	sparse implementation, memory
prediction over interpretation	targets generalization, not zeros	final metric and operating cost

4 Lasso

4.1 The L_1 penalty sets some coefficients exactly to zero

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1\Big], \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

It pushes not only large but also small coefficients to 0, giving a **sparse** model: a short variable list together with predictions. What is *not* gained: one cannot conclude that unselected variables are truly irrelevant.

Theorem: soft-thresholding under an orthogonal design

If $X^\top X = nI$, let $z = X^\top \mathbf{y}/n$ (the OLS estimate). Then

$$\hat{\beta}_j^{\text{Lasso}} = S(z_j, \lambda/2) = \text{sign}(z_j)(|z_j| - \lambda/2)_+, \qquad \hat{\beta}_j^{\text{Ridge}} = \frac{z_j}{1+\lambda}.$$

Proof. $\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \frac{1}{n}\|\mathbf{y}\|^2 - 2\boldsymbol{\beta}^\top z + \|\boldsymbol{\beta}\|^2$, so the Lasso objective separates into $\sum_j\{(\beta_j - z_j)^2 + \lambda|\beta_j|\}$ up to a constant. For one coordinate the subgradient condition is $2(\beta_j - z_j) + \lambda s_j = 0$ with $s_j = \text{sign}(\beta_j)$ if $\beta_j \neq 0$ and $s_j \in [-1, 1]$ if $\beta_j = 0$. If $|z_j| \leq \lambda/2$, $\beta_j = 0$ satisfies it; otherwise $\beta_j = z_j - \frac{\lambda}{2}\text{sign}(z_j)$. For Ridge, $(\beta_j - z_j)^2 + \lambda\beta_j^2$ is minimized at $z_j/(1+\lambda)$. ■

Ridge shrinks proportionally and never reaches zero; Lasso subtracts a constant and sets small coefficients exactly to zero. This is the algebraic reason for sparsity; in general designs, coordinate descent applies exactly this soft-threshold to partial residuals one coordinate at a time.

4.2 Geometry: the ball and the diamond

In the constrained form the elliptical contours of the squared loss meet the constraint region. The Ridge ball $\|\boldsymbol{\beta}\|_2^2 \leq c$ has a smooth boundary, so the contact point is typically off the axes. The Lasso diamond $\|\boldsymbol{\beta}\|_1 \leq c$ has corners on the axes, so contact often happens at a corner, where some coefficients are 0. The coefficient paths show this: under Ridge many coefficients shrink together; under Lasso some disappear exactly.

4.3 Lasso is unstable among correlated variables

If several variables measure the same latent factor, Lasso may select one and set another to 0; with a slightly different sample a different name is selected. That a variable was not selected does not mean its predictive information is 0, and selection results must not be over-interpreted causally. Check selection frequencies across resamples or CV folds and whether variables are domain-equivalent; with many correlated signals keep Ridge or Elastic Net as candidates.

Remark: a limit on Lasso selection

When $p > n$, the Lasso selects at most n variables. For two identical columns the Lasso solution is not unique: any split (β_1, β_2) with the same sum and signs is optimal.

Situation	Why Lasso is attractive	Caveat
variables costly to measure or deploy	simpler prediction pipeline	compare selection frequency and loss in performance
very many variables, sparse signal suspected	shrinks unnecessary coefficients to 0	may keep only one of correlated signals
an explainable operating model is needed	easy to present a variable list	do not turn selection into causal claims
screening for later analysis	starting point of exploratory reduction	needs separate validation and domain review

5 Elastic Net

$$\hat{\boldsymbol{\beta}}^{\mathrm{EN}} = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\{\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\}\Big], \qquad \alpha \in [0,1].$$

$\alpha = 1$ is Lasso, $\alpha = 0$ is Ridge. The L_2 part makes the objective strictly convex, so the solution is unique, and gives a **grouping effect**: highly correlated variables receive similar coefficients and tend to enter or leave together. The L_1 part keeps sparsity. Under an orthogonal design the solution is $S(z_j, \lambda\alpha/2)/\{1 + \lambda(1-\alpha)\}$: soft-thresholding followed by proportional shrinkage. Both λ and α are tuned by CV.

6 Choosing λ Inside the Folds

Within each fold, standardization, transformations and fitting of each λ candidate must depend only on the training part; only then is the validation score fair. The CV curve gives two common rules:

- **minimum rule**: choose $\lambda_{\min}$ with the smallest mean CV loss, natural when predictive performance is all that matters;
- **1-SE rule**: $\lambda_{1\mathrm{SE}} = \max\{\lambda : \widehat{R}_{\mathrm{CV}}(\lambda) \le \widehat{R}_{\min} + \mathrm{SE}_{\min}\}$, the simplest (most regularized) model within one standard error of the minimum; useful when interpretability, computation and stability matter.

The final selection rule is fixed *before* the analysis. The lowest point of the CV curve is not the only answer: differences within the noise level of CV are not meaningful.

7 Regularized Logistic Regression

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\sum_{i=1}^{n} \ell\{y_i, \sigma(\beta_0 + x_i^\top \boldsymbol{\beta})\} + \lambda P(\boldsymbol{\beta})\Big], \qquad \ell(y,p) = -[y\log p + (1-y)\log(1-p)].$$

What changes: log loss replaces squared error. What stays: standardization, L_1/L_2 penalties, CV selection of λ, protection of the test set. The penalty also fixes the separation problem of Ch. 1: with $\lambda > 0$ the objective is coercive, so a finite minimizer exists.

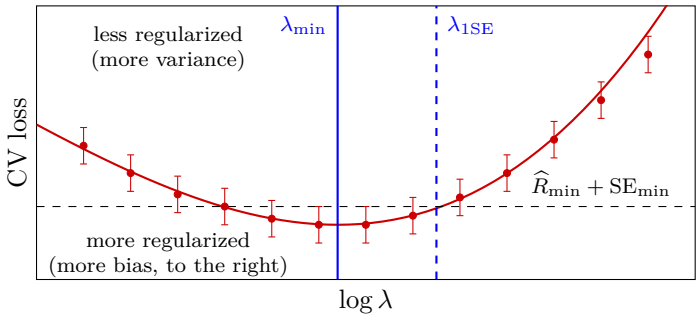


Figure 1: A typical CV curve with ±1 SE bars (schematic). The minimum rule takes $\lambda_{\min}$; the 1-SE rule moves right to the largest λ whose CV loss is still below the dashed line, giving a simpler, more stable model at almost the same estimated risk.

8 Regularization Versus Robust Regression

Regularization	Robust regression
reduces coefficient variance and overfitting caused by many variables, collinearity and large degrees of freedom	mitigates loss domination caused by a few large residuals, label errors and extreme cases
Ridge, Lasso, Elastic Net	Huber, quantile, LAD
shrinkage, selection, generalization risk	influence of outliers, conditional location

They address *different failure causes* and can be combined (e.g. Huber loss with an Elastic Net penalty).

9 Practical Selection and Reporting

Prediction-centered: prefer the lowest CV loss, but if differences are within the SE choose the simpler or faster candidate. **Operation-centered**: if the loss in performance is small prefer fewer variables, more stable selection and easier retraining. **Interpretation-centered**: look at signs, selection frequency and domain constraints together; one Lasso fit does not settle variable importance. **Final step**: fix the selection rule, refit on all training data, evaluate the protected test set once.

Example: reading a results table (illustrative)

Candidate	CV primary metric	Complexity	Interpretation
OLS	0.842 ± 0.031	180 variables	baseline; unstable under collinearity
Ridge	0.801 ± 0.019	180 shrunk coefficients	better performance and stability
Lasso	0.808 ± 0.026	24 selected	small loss in performance for simplicity
Elastic Net	0.799 ± 0.020	47 selected	compromise for correlated blocks

(Loss: smaller is better.) Do not copy the numbers; adopt the format that reads mean, variability, complexity and operating cost together.

Example: customer churn case

Situation. 8,000 customers, 600 derived variables; strong correlations among usage, payment and inquiry variables; churn rate 8%. **Candidates.** Standardized logistic baseline, Ridge, Lasso, Elastic Net. Primary metrics PR-AUC and log loss; operating metric recall@budget. **Split.** Protect the test period by time order; stratified CV within the training period. **Report.** CV mean and variability, chosen λ, number of selected variables, probability calibration and test performance in one table.

Summary

1. Many variables and collinearity make *coefficient variance and generalization risk*, rather than training fit, the main problem of OLS.
2. Ridge shrinks, Lasso selects, Elastic Net compromises; the right choice depends on the signal structure and the operating goal.
3. Standardization and λ selection are done inside CV folds; regularization does not automatically fix leakage or outliers.